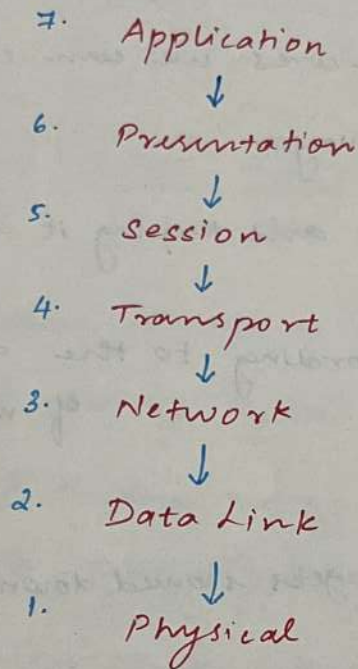


Day-6: OSI LAYERS

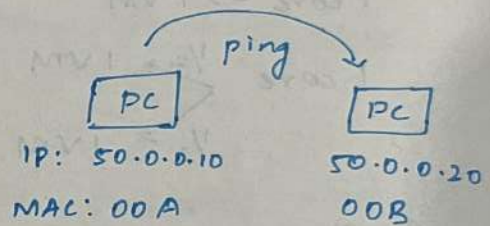
- It is a standard used in all PCs.
- There are seven layers totally.

OSI - Open system interconnection



APPLICATION:

Echo



When we ping a device to check network reachability, we will get a reply dummy data.

- That is called as Echo.
- This data generated in Application layer has 32 bytes.
- ICMP is a protocol that carries the data across all the layers.

ICMP - Internet control message protocol.

- Data gets in and out of the layers through physical layer.

PRESENTATION:

Echo	F
------	---

→ Adds an encapsulation to know the format of the data which is generated in the app. layer.

SESSION:

Echo	F	Time
------	---	------

→ This is the timing encapsulation.

→ It determines the sending & receiving time.

TRANSPORT:

Echo	F	Time	
------	---	------	--

→ This layer also adds an encapsulation.

NETWORK:

Echo	F	Time		SOURCE IP: 50.10 DEST. IP: 50.20
------	---	------	--	-------------------------------------

→ Internet protocol adds the source IP & destination IP in the network layer.

→ When the source & dest. IP are added it is called as packet encapsulation.

DATA LINK:

Echo	F	Time		SIP: 50.10 DIP: 50.20	Source mac: 00A Dest. mac: 00B
------	---	------	--	--------------------------	-----------------------------------

→ A PC can only know its source MAC address and not the destination MAC address because while pinging we are giving only the dest. IP address.

→ This can be solved by using Ethernet Protocol

→ Another protocol ARP is used to find the ^{mac} address of dest. PC.

→ Frame encapsulation

Address Resolution Protocol (ARP): [Layer 2 protocol]

It is a protocol used to find the destination mac with help of destination ip.

ARP REQUEST

I am: 50.10
My mac: 00A
Who is: 50.20
Give your mac

Broadcast msg. sent
to all devices.

But device with
dest. IP will only reply

SM: 00A

DM: FFFF (12 F)

↓

It sends a broadcast
message to all the devices
in connected network.

ARP REPLY

I am: 50.20
My mac: 00B
Your IP: 50.10
Your mac: 00A

SM: 00B

DM: 00A.

↓

This is sent unicast
because we already
know the mac address
in request.

ARP TABLE

50.0.0.20 → 00B

This table has the value that we
found for 5 minutes. In that
5 minutes if we ping again all
process repeats but ARP checks the
table once & gives value directly.

→ If source & dest. MAC are given as
encapsulation then it is called as **Frame encapsulation**.
Then it goes to physical layer.

PHYSICAL:

→ NIC: Binary → Electrical signal

Electrical → Binary

protocol]

How many packets are sent & received when we ping first time?

- 1) $\longrightarrow$ ARP REQUEST
- 2) $\longleftarrow$ ARP REPLY
- 3) $\longrightarrow$ ICMP REQ.
- 4) $\longleftarrow$ ICMP REP.
- 5) $\longrightarrow$ ICMP REQ.
- 6) $\longleftarrow$ ICMP REP.
- 7) $\longrightarrow$ ICMP REQ.
- 8) $\longleftarrow$ ICMP REP.
- 9) $\longrightarrow$ ICMP REQ.
- 10) $\longleftarrow$ ICMP REP.

Total - 10 packets

If we ping second time only 8 packets will be generated because ARP Req. & Rep. not needed.

[ping 192.168.1.2 -t] $\rightarrow$ If we give this command then continuously reply will be sent until we stop.